



MECH MAINT- OFF SITE Department  
SIMHADRI

**Note No :1**



**Ref:**SMTTP/MECH MAINT-/2024-25/LMI /1041198

**Date:**14/03/2025(17:42:35)

**Subject:**Review of LMI “FIRE PROTECTION OF POWER STATIONS”

**Subject: Review of LMI “FIRE PROTECTION OF POWER STATIONS”**

**LMI No: LMI/OGN/OPS/SYST/002/**

The above approved LMI was reviewed in the month of Aug.’2022. The changes in LMI are incorporated as per details at revision index of LMI.

Put up for approval please.

**Submitted by:**

राहुल जैन - उप महाप्रबंधक

Rahul Jain - 013328 (DY. GENERAL MANAGER ,MECH MAINT- OFF SITE) ,Simhadri  
(RAHULJAIN@NTPC.CO.IN, 9415246668)

**On:**14/03/2025(17:42:35) **Ref:**SMTTP/MECH MAINT-/2024-25/LMI /1041198

**Note No:2**

Revised LMI is put up for approval by competent authority please.

**Submitted by:**

टी वी राजशेखर - अपर महाप्रबंधक

T V Rajasekhar - 007005 (ADDL. GENERAL MANAGER ,MECH MAINT- OFF SITE)  
(TVRSEKHAR@NTPC.CO.IN, 9650998176) ,Simhadri

**On:**14/03/2025(18:27:21) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:3**

It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN

**Submitted by:**

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)  
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

**On:**15/03/2025(10:05:38) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:4**

**Submitted by:**

राहुल जैन - उप महाप्रबंधक

Rahul Jain - 013328 (DY. GENERAL MANAGER ,MECH MAINT- OFF SITE)  
(RAHULJAIN@NTPC.CO.IN, 9415246668) ,Simhadri

**On:**21/03/2025(18:04:42) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:5**

please prepare LMI as per standard format

**Submitted by:**

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)  
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

**On:**22/03/2025(10:49:50) **Ref:**SMTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:6**

Please prepare the LMI as per observation of EEMG (Note 5 on NP-3).

**Submitted by:**

महेश अलुगोलु - अपर महाप्रबंधक

Mahesh Alugolu - 006936 (ADDL. GENERAL MANAGER ,CIVIL MAINT.)  
(AMAHESH@NTPC.CO.IN, 9650993435) ,Simhadri

**On:**22/03/2025(11:50:16) **Ref:**SMTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:7**

Necessary done.

**Submitted by:**

राहुल जैन - उप महाप्रबंधक

Rahul Jain - 013328 (DY. GENERAL MANAGER ,MECH MAINT- OFF SITE)  
(RAHULJAIN@NTPC.CO.IN, 9415246668) ,Simhadri

**On:**27/03/2025(18:19:39) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:8**

**Submitted by:**

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)  
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

**On:**28/03/2025(16:52:07) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:9**

**Submitted by:**

राहुल जैन - उप महाप्रबंधक

Rahul Jain - 013328 (DY. GENERAL MANAGER ,MECH MAINT- OFF SITE)  
(RAHULJAIN@NTPC.CO.IN, 9415246668) ,Simhadri

**On:**28/03/2025(18:21:30) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:10**

**Submitted by:**

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)  
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

**On:**29/03/2025(17:38:43) **Ref:**SMTTP/MECH MAINT-/2024-25/LMI /1041198

**Note No:11**

**Submitted by:**

राहुल जैन - उप महाप्रबंधक

Rahul Jain - 013328 (DY. GENERAL MANAGER ,MECH MAINT- OFF SITE)  
(RAHULJAIN@NTPC.CO.IN, 9415246668) ,Simhadri

**On:**03/04/2025(11:26:23) **Ref:**SMTTP/MECH MAINT-/2024-25/LMI /1041198

**Note No:12**

LMI is prepared as per OGN/OPS/SYST/002 as per Simhadri LMI format

**Submitted by:**

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)  
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

**On:**04/04/2025(11:39:54) **Ref:**SMTTP/MECH MAINT-/2024-25/LMI /1041198

**Note No:13**

As discussed, please check.

**Submitted by:**

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)  
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

**On:**04/04/2025(11:59:05) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:14**

**Submitted by:**

राहुल जैन - उप महाप्रबंधक

Rahul Jain - 013328 (DY. GENERAL MANAGER ,MECH MAINT- OFF SITE)  
(RAHULJAIN@NTPC.CO.IN, 9415246668) ,Simhadri

**On:**18/04/2025(16:19:35) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:15**

**Submitted by:**

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)  
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

**On:**19/04/2025(10:07:54) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:16**

**Submitted by:**

राहुल जैन - उप महाप्रबंधक

Rahul Jain - 013328 (DY. GENERAL MANAGER ,MECH MAINT- OFF SITE)  
(RAHULJAIN@NTPC.CO.IN, 9415246668) ,Simhadri

**On:**19/04/2025(11:02:41) **Ref:**SMTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:17**

LMI is prepared as per OGN ,may be approved

**Submitted by:**

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)  
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

**On:**22/04/2025(17:32:54) **Ref:**SMTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:18**

**Submitted by:**

एम सी यादव - उप महाप्रबंधक

Madhup Chandra Yadav - 057675 (DY. GENERAL MANAGER ,CIVIL MAINT.)  
(MCYADAV01@NTPC.CO.IN, 9319615566) ,Simhadri

**On:**23/04/2025(10:07:08) **Ref:**SMTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:19**

Forwarded for further process.

**Submitted by:**

चन्द्र शेखर गोडी - उप महाप्रबंधक

CHANDRA SEKHAR GODI - 085565 (DY. GENERAL MANAGER ,CIVIL MAINT.)  
(CSGODI@NTPC.CO.IN, 8332862038) ,Simhadri

**On:**28/04/2025(15:54:57) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:20**

**Submitted by:**

पी गंगाधर - अपर महाप्रबंधक

Padalahariveera Gangadhar - 008164 (ADDL. GENERAL MANAGER ,CIVIL MAINT.)  
(PHVGANGADHAR@NTPC.CO.IN, 9650998462) ,Simhadri

**On:**07/05/2025(17:15:58) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:21**

**Submitted by:**

आन्नप्रगद रमा कोटेश्वरराव - अपर महाप्रबंधक

Annapragada Rama Koteswararao - 047211 (ADDL. GENERAL MANAGER ,FUEL MANAGEMENT)  
(ARKOTESWARARAO@NTPC.CO.IN, 8332988907) ,Simhadri

**On:**07/05/2025(17:33:54) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198



**Note No:22**

Checked and forwarded.

**Submitted by:**

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

Somnath Bhattacharya - 007203 (ADDL. GENERAL MANAGER ,OPERATION)  
(SOMNATHBHATTACHARYA@NTPC.CO.IN, 9431600725) ,Simhadri

**On:**10/05/2025(20:57:31) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:23**

**Submitted by:**

जे आर वर्गीस - अपर महाप्रबंधक

Jayamol Rachel Varughese - 006127 (ADDL. GENERAL MANAGER ,MTP)  
(JAYAMOLRV@NTPC.CO.IN, 9440918557) ,Simhadri

**On:**12/05/2025(09:58:46) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:24**

Forwardede for further processing.

**Submitted by:**

अजय गिरि - अपर महाप्रबंधक

Ajay Giri - 007629 (ADDL. GENERAL MANAGER ,FGD)  
(AJAYGIRI@NTPC.CO.IN, 7440228332) ,Simhadri

**On:**14/05/2025(15:52:04) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:25**

**Submitted by:**

राजेश चावला - अपर महाप्रबंधक

Rajesh Chawla - 007383 (ADDL. GENERAL MANAGER ,ASH HANDLING)  
(RAJESHCHAWLA@NTPC.CO.IN, 9425534528) ,Simhadri

**On:**15/05/2025(16:37:45) **Ref:**SMTTP/MECH MAINT-/2024-25/LMI /1041198

**Note No:26**

Please review the LMI

**Submitted by:**

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE)  
(NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri

**On:**16/05/2025(11:21:36) **Ref:**SMTTP/MECH MAINT-/2024-25/LMI /1041198

**Note No:27**

Reviewed and forwarded for further processing.

**Submitted by:**

के पी रमेश - वरिष्ठ प्रबंधक

KANADE PIYUSH RAMESH - 101897 (SENIOR MANAGER ,MECH MAINT- TURBINE)  
(PIYUSHKANADE@NTPC.CO.IN, 9425219272) ,Simhadri

**On:**17/05/2025(10:05:15) **Ref:**SMTTP/MECH MAINT-/2024-25/LMI /1041198

**Note No:28**

Checked and forwarded for further processing.

**Submitted by:**

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE)  
(NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri

**On:**19/05/2025(10:42:56) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:29**

**S.No-4.5, Column-Fire prevention point-2** matter related to Feeder water spray nozzle can be removed as this system is not provided at Simhadri.

**S.No-4.7, Columns-Fire hazard & Fire prevention point-1** related matter can be removed as at Simhadri LDO at ambient temperature is in use. Also for oil atomization air is used in place of steam.

**S.No-4.8, Column-Fire prevention point-4** related matter can be removed as at Simhadri LDO is in use, now cooler is not required.

**S.No-4.11, Coumn-Resp,** HOD-MM BMD may be removed as the area relevant is Turbine Generator.

**Submitted by:**

सुरेश बाबु कोल्ला - अपर महाप्रबंधक

Suresh Babu Kolla - 007980 (ADDL. GENERAL MANAGER ,MECH MAINT- BOILER)  
(SBKOLLA@NTPC.CO.IN, 9490307556) ,Simhadri

**On:**19/05/2025(14:10:16) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:30**

Necessary changes done.

Submitted please

**Submitted by:**

राहुल जैन - उप महाप्रबंधक

Rahul Jain - 013328 (DY. GENERAL MANAGER ,MECH MAINT- OFF SITE)  
(RAHULJAIN@NTPC.CO.IN, 9415246668) ,Simhadri

**On:**19/05/2025(16:15:34) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:31**

The revised LMI has been reviewed and found in line with the relevant OGN document.

**Submitted by:**

सुरेश बाबु कोल्ला - अपर महाप्रबंधक

Suresh Babu Kolla - 007980 (ADDL. GENERAL MANAGER ,MECH MAINT- BOILER)  
(SBKOLLA@NTPC.CO.IN, 9490307556) ,Simhadri

**On:**19/05/2025(20:09:07) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:32**

Reviewed and submitted for approval.

**Submitted by:**

ओम प्रकाश - अपर महाप्रबंधक

Om Prakash - 007105 (ADDL. GENERAL MANAGER ,MAINTENANCE)  
(OMPRAKASH08@NTPC.CO.IN, 9650994070) ,Simhadri

**On:**20/05/2025(10:02:43) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:33**

Revised LMI is in line with the OGN (FIRE PROTECTION OF NTPC STATIONS).

LMI may please be approved.

**Submitted by:**

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)  
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

**On:**20/05/2025(10:16:05) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI /1041198

**Note No:34**

**Submitted by:**

प्रसन्ता कुमार जेना - महाप्रबंधक

Prasanta Kumar Jena - 006081 (GENERAL MANAGER ,O & M)  
(PKJENA@NTPC.CO.IN, 9650990836) ,Simhadri

**On:**20/05/2025(15:57:10) **Ref:**SMTTP/MECH MAINT-/2024-25/LMI /1041198

**Note No:35**

**Approved & Submitted by:**

समीर शर्मा - कार्यकारी निदेशक

Sameer Sharma - 005041 (EXECUTIVE DIRECTOR ,HEAD OF PROJECT)  
(SAMEERSHARMA@NTPC.CO.IN, 9650990022) ,Simhadri

**On:**20/05/2025(20:24:15) **Ref:**SMTTP/MECH MAINT-/2024-25/LMI /1041198

**LOCATION MANAGEMENT INSTRUCTION**

**FIRE PROTECTION OF POWER STATIONS**

**LMI/OGN/OPS/SYST/002/REVISION NO:04**

**March 2025**

**Approved for Implementation by ED (SIMHADRI)**

## LMI APPROVAL DOCUMENTS

|                        |                                                              |
|------------------------|--------------------------------------------------------------|
| 1 LMI Name             | FIRE PROTECTION OF POWER STATIONS                            |
| 2. LMI No.             | LMI/OGN/OPS/SYST/002/ Rev:04                                 |
| 3. Reference Document  | OGN/OPS/SYST/002/Rev-0, April 2001                           |
| 4. Superseded Document | LMI/OGN/OPS/SYST/002/ Rev:03                                 |
| 5.Prepared By          | NAME: Rahul Jain<br>DGM/MMOFS<br>SIGNATURE:                  |
| 6.Checked By           | NAME: T V Rajasekhar<br>AGM/MMOFS<br>SIGNATURE:              |
| 6a) Checked By         | NAME: Annapragada Rama Koteswararao<br>AGM/CHP<br>SIGNATURE: |
| 6b) Checked By         | NAME: Somnath Bhattacharya<br>AGM/OPERATION<br>SIGNATURE:    |
| 6c) Checked By         | NAME: Malay Kanti Jana<br>AGM/EMD<br>SIGNATURE:              |
| 6d) Checked By         | NAME: Ajay Giri<br>AGM/C& I Malay Kanti Jana<br>SIGNATURE:   |
| 6e) Checked By         | NAME: Suresh Babu Kolla<br>AGM/BMD<br>SIGNATURE:             |
| 7.Rechecked By         | NAME: Om Prakash<br>AGM/Maintenance<br>SIGNATURE:            |
| 8.Reviewed By          | NAME:Prasanta Kumar Jena<br>GM ( O & M)<br>SIGNATURE:        |
| 9. Approved by         | NAME: Sameer Sharma<br>ED (SIMHADRI)<br>SIGNATURE:           |



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|         |                                          |
|---------|------------------------------------------|
| TITLE   | <b>FIRE PROTECTION OF POWER STATIONS</b> |
| LMI No. | LMI/OGN/OPS/SYST/002/ Rev:04             |

**1. INTRODUCTION:**

Power plant is exposed to onerous operating conditions and the combination of flammable material in large quantities and ignition sources which can lead to the incidences of fire. At Simhadri various fire detection systems viz. Quartzoid bulb type heat detectors, linear heat sensing cable type heat detectors ,infra-red type head detectors ,ionization and photo electric type smoke detectors and various fire protection system viz. hydrant system ,portable fire extinguishers ,automatic medium velocity and high velocity spray systems ,automatic inert gas extinguishing system, sprinkler system, foam injection system is installed at various locations.

The objective of this LMI is to provide guidelines to operation and maintenance staff for fire prevention vis-à-vis various hazard situations and the checks/maintenance jobs to be carried for ensuring reliability of the fire detection and protection systems.

**2.SUPERSEDED DOCUMENT:** LMI/OGN/OPS/SYST/002/Rev 3.00

**3. Details of Revision:** It is mentioned in the Revision Index.

**3.1SCOPE:**

The scope of this LMI is to detail the methods that are to be adopted for fire prevention vis-à-vis various hazard situations and the checks/maintenance jobs to be carried for ensuring reliability of fire detection and protection systems. The scope of the LMI is limited to fixed fire detection and protection systems.

Recommendations are made for prevention/detection/protection against fire, for inspection, for testing & maintenance and for operational needs pertaining to the fire detection and protection equipment

**3.2REFERENCE DOCUMENT**

Operation guidance note OGN/OPS/SYST/002 circulated by operation services-NTPC corporate center.

**3.3. SPECIFIC INSTRUCTIONS**

To achieve the objective of the LMI the following approach is to be followed:

- a) Identification of fire hazards and required preventive action corresponding to various locations
- b) Inspection, testing and maintenance requirements for various fire detection and protection systems

**4.0 FIRE HAZARDS AND PREVENTION**

The various hazards in a power station and the methods to prevent these are given in the following table along with responsibility centers.

|         |                                          |
|---------|------------------------------------------|
| TITLE   | <b>FIRE PROTECTION OF POWER STATIONS</b> |
| LMI No. | LMI/OGN/OPS/SYST/002/ Rev:04             |

#### 4.0 **METHODOLOGY OF IMPLEMENTATION AND RESPONSIBILITY CENTERS**

| S.NO | Area                                      | Fire Hazard                                                                                                                                                                      | Fire Prevention                                                                                                                                                                                                                                                                                                                                                         | Responsibility |
|------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 4.1  | Coal Stockyard & <b>Stacker Reclaimer</b> | Self-ignition due to coal, air interaction (Dry coal-wet air during March to June period is the most critical combination and moist coal-dry air the most favorable combination) | 1. Coal to be wet while stacking.<br>2. Coal to be stacked layer by layer (each of 1-1.5M height) with compaction of each layer.<br>3. Regular spray of water over coal yard to be done.<br>4. Proper records of period of stacking to be maintained and the principle of FIFO (First In First Out) to be adopted.<br>5. Fresh coal is not to be stacked over old coal. | HOD(FM)        |
| 4.2  | Bulldozers, Pay-loaders etc.              | Coal dust and oil deposits combined with hot exhaust pipes, fuel oil tank, fuel oil leakages, cables                                                                             | Regular cleaning of engine compartment, inspection to check for any fuel leakage.                                                                                                                                                                                                                                                                                       | HOD(FM)        |

| TITLE   |                | FIRE PROTECTION OF POWER STATIONS                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|---------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| LMI No. |                | LMI/OGN/OPS/SYST/002/ Rev:04                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
| S.NO    | Area           | Fire Hazard                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Fire Prevention                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Resp.    |
| 4.3     | Coal Conveyors | 1. Maintenance activities involving welding / gas cutting leading to weld spatter on coal/ conveyor.<br>2. Friction caused by<br>a) Belt with steel frame<br>b) Jammed idler<br>c) Jammed conveyor belt due to choking of chute but pulley rotating.<br>3. Fluid coupling failures<br>4. Overfilled gear box<br>5. Self-ignition of accumulated coal dust at concave contours of conveyors.<br>6. Loosely laid wires / cables / welding cables with exposed joints | 1. Regular cleaning and inspection of all areas of CHP.<br>2. Chute block switches to be kept in service to alarm chute choking.<br>3. Pulley, idlers, gear boxes, fluid couplings not to be over greased/over filled.<br>4. System of hot work permit to be followed during welding / cutting job.<br>5. Only fire retardant (FR grade) belt to be used including bunker sealing belt, ILMS belt.<br>6. Belt not to rub steel structure.<br>7. While welding any chute/hopper lined with rubber / polymer liners, special care to be taken to cool the liner.<br>8. Adequate illumination in all areas of CHP.<br>9. Coal dust accumulation on magnets and its cleaning belt to be avoided as temperatures are high there.<br>10. Loose wires / cable/welding leads with open joints to be avoided. Belt flapping to be avoided by installing rollers mounted on carrying side.<br>11. Fire Blanket to be used in case welding/cutting on Belt | HOD (FM) |

| TITLE   |              | FIRE PROTECTION OF POWER STATIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                       |
|---------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LMI No. |              | LMI/OGN/OPS/SYST/002/ Rev:04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                       |
| S.NO    | Area         | Fire Hazard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Fire Prevention                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Resp.                                                                                                                                                                 |
| 4.4     | COAL BUNKERS | <ol style="list-style-type: none"> <li>1. Feeding of hot coal from stockyard which may be about to burn.</li> <li>2. Residual coal stacking to bunker walls for a long periods resulting in self- ignition.</li> <li>3. Long period coal handling in bunkers particularly top mills.</li> <li>4. Any draught of air through the coal in bunkers, particularly compressed air from air blast equipment.</li> <li>5. Low bunker level coupled with unsatisfactory isolation from feeder/mill with subsequent pressurization of furnace causing hot gases to pass through the bunker.</li> <li>6. Bunker conveyor fires</li> </ol> | <ol style="list-style-type: none"> <li>1. Non- moving coal should be avoided in bunkers should be emptied and cleaned thoroughly during overhaul of unit.</li> <li>2. Bunker gates to be maintained to ensure that they close properly when bunkers are not to be use .</li> <li>3. Operation must do mill change overs so that no bunker old coal for long durations.</li> <li>4. Hot coal should not be fed into the bunkers from stockyard.</li> <li>5. System of hot work permits to be followed during welding/cutting job in a bunker floor</li> <li>6. Availability of bunker level indication along with annunciation for low bunkers level to be ensured.</li> </ol> | <p>HOD (FM),HOD (BMD), HOD (OPRN)</p> <p>HOD (OPRN),HOD MM-BMD)&amp;HOD (EM)</p> <p>HOD (OPRN)</p> <p>HOD (FM)</p> <p>HOD(FM)</p> <p>HOD (C&amp;I) &amp; HOD (FM)</p> |

| TITLE   |                             | FIRE PROTECTION OF POWER STATIONS                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                           |
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| LMI No. |                             | LMI/OGN/OPS/SYST/002/ Rev:04                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                           |
| S.N     | AREA                        | Fire Hazard                                                                                                                                                                                                                                  | Fire Prevention                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Resp.                                                                                     |
| 4.5     | Coal Feeders                | 1. Hot or burning coal passing from coal bunkers.<br>2. Fires in mills particularly during start up or shut down periods.<br>3. Friction due to mechanical failure<br>4. Welding /gas cutting etc.<br>5. Residual coal in pockets of feeder. | 1. Routine cleaning of feeder from inside to reduce extent of residual coal left compacted in dead spaces within the feeder.<br><b>2. Feeder inlet and outlet gate should be in healthy condition. Coal feeder should be kept empty when not in service. Mill inerting should be in healthy condition. Inerting Valve should operate from control room in case of biomass firing.</b><br>3. Bunker gates and mills dampers to be maintained to prevent the passage into the feeder of combustion products from fires elsewhere in the system.<br>4. Periodic checking of feeders protection on "No Coal on belt" to be ensure.<br>5. Ensure feeder emptied, & feeder inlet gate closed. scrapper conveyer area cleaned.<br>6. Ensure cleaning of the area before work is in progress.<br>7. The system of hot work permits to be strictly followed during welding / gas cutting jobs. | HOD (MM-BMD)<br><br>HOD (MM-BMD)<br><br>HOD (C&I)<br><br>HOD (MM-BMD)<br><br>HOD (MM-BMD) |
| 4.6     | Coal Mills and P.F. Piping. | High mill outlet temperature due to coal from interruption.                                                                                                                                                                                  | 1. Proper start-up and shut down procedure to be maintained.<br>2. Auto closing of HAG from mill outlet temperature higher protection is to be kept in service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | HOD (OPRN)                                                                                |

| TITLE   |                            | FIRE PROTECTION OF POWER STATIONS                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |
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| LMI No. |                            | LMI/OGN/OPS/SYST/002/ Rev:04                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |
| S.N     | AREA                       | Fire Hazard                                                                                                                                                                                                      | Fire Prevention                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Resp.                                                                                                     |
| 4.7     | Boiler<br>Burner<br>Fronts | 1.Rupture of flexible hoses.<br>2.Leakage of P.F pipes.<br><br>3.Faulty igniters coupled with leaking oil.<br><br>4.Scrap material<br><br>5. JB near boiler burners                                              | Prompt attention to defects on burner connecting hoses and pipe work.<br>No spillage of oil around burner area during removal of burner carrier for tip cleaning. In case of oil leakage, soaked insulation to be removed and re-insulation to be done.<br>Igniter faults to be immediately attended.<br>Periodic checking of igniter healthiness in all respects to be done.<br>Oil waste, rag, and combustible materials no to be allowed to accumulate.<br>Minimum electrical equipment / cables / junction boxes near burners to be located. | HOD(MM-BMD)<br><br>HOD(MM-BMD)<br><br>HOD(MM-BMD) & HOD(C&I)<br>HOD(C&I)<br><br>HOD(OPRN)<br><br>HOD(EMD) |
| 4.8     | Fuel oil storage tanks     | 1.Greater risk during venting, gassing and filling operations of light fuel oil.<br>2. Leakage of tank.<br>3. Dry grass around tanks.<br>4.Unclean surroundings<br>5.Soaked insulation.<br>6.Electrial Earthing. | 1. Precautions to be taken during tank emptying, cleaning, tank inspection, vent inspections, other maintenance works ( including electrical/lighting works).<br>2. Entry to bund area to be restricted to authorized persons only.<br>3.All areas within the bund walls must be free of vegetation and other combustible materials.<br>4. Surroundings to be kept clean.<br>5.Oil-soaked insulation to be removed.<br>6. Earthing to be checked.                                                                                                | HOD (MM-BMD)<br><br>HOD (OPRN)<br>HOD (MM-BMD)<br>HOD (OPRN)<br>HOD (MM-BMD)<br><br>HOD (EM)              |

|     |                                            |                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                          |
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|     |                                            | NTPC                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SIMHADRI                                                                                                 |
|     | TITLE                                      | FIRE PROTECTION OF POWER STATIONS                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                          |
|     | LMI No.                                    | LMI/OGN/OPS/SYST/002/ Rev:04                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                          |
| 4.9 | Fuel oil unloading and transfer pump house | 1. Oil spillage or leakage due to failure of joints or of venting due to overheating or restrictions in system.<br>2. Heaters<br>3. Loosely laid electrical cables/ welding cables.<br>4. Drain oil stagnant in unloading pipe tranches and surface drains. | 1. Adequate ventilation in the enclosed pump house to keep the concentration of oil vapors within safe levels.<br>2. Proper earthing of all electrical systems.<br>3. Use of flame proof lighting / electrical appliances only in the pump house.<br>4. Trenches to be cleaned of oil and debris.<br>5. Oil waste, rag, other combustible materials, not to be allowed to accumulate.<br>6. No spillage of oil.<br>7. Prompt attention to defect on oil pipes heaters etc. | HOD (OPRN)<br>HOD (OPRN)<br>HOD (EM)<br><br>HOD (EM)<br>HOD (OPRN)<br><br>HOD(MM-BMD)<br><br>HOD(MM-BMD) |



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|---------|------------------------------------------|
| TITLE   | <b>FIRE PROTECTION OF POWER STATIONS</b> |
| LMI No. | LMI/OGN/OPS/SYST/002/ Rev:04             |

| S.N  | AREA                       | Fire Hazard                                                                                                                                                                                                          | Fire Prevention                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Resp.                                                                                              |
|------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 4.10 | Regenerative air preheater | <p>1. Carryover of unburnt liquid fuel onto the internal heat exchanger surfaces during cold start up.</p> <p>2. Slow build- up of deposits of unburnt fuel in the absence of soot blowing for a period of time.</p> | <p>1. Inspection of air heater during shutdown.</p> <p>2. Soot blowing of air heaters especially during start up immediately after bringing boiler into service and cleaning regularly.</p> <p>3. Oil to be burnt with maximum attainable efficiency in the furnace by appropriate cleaning &amp; adjustment of burners, to minimize the quantity of unburnt particles, elimination of tramp air and consequent chilling of the flame and by ensuring that combustion chamber are gas tight.</p> <p>4. Regular cleaning, maintenance and testing of oil burners.</p> <p>5. Thermo- couple type fire detection system to be kept in service for monitoring air heater temperatures.</p> | <p>HOD (MM-BMD)</p> <p>HOD (OPRN)</p> <p>HOD (MM-BMD)</p> <p>HOD (MM-BMD)</p> <p>HOD (C&amp;I)</p> |

| NTPC    |                                 | SIMHADRI                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                      |  |
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| TITLE   |                                 | FIRE PROTECTION OF POWER STATIONS                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                      |  |
| LMI No. |                                 | LMI/OGN/OPS/SYST/002/ Rev:04                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                      |  |
| S.N     | AREA                            | Fire Hazard                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Fire Prevention                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Resp.                                                                                                                                                                |  |
| 4.11    | Turbine generator               | 1. Minor leakages of lubricating and control fluids into inaccessible area, lagging, drain trenches and coming in contact with hot surfaces.<br>2. Heavy oil leakage in the form of spray on to bare hot metal caused by bursting of pipe / flange.<br>3. Leakage of hydrogen to atmosphere.<br>4. Leakage of hydrogen into bus duct coupled with choking of vent of bus duct.<br>5. Cables laid near hot equipments.<br>6. Looseness of current carrying bolts carbon brushes. | 1. Oil leaks if any, should be collected nearest to the leak.<br>2. Prompt attention to oil leakages.<br>3. Good housekeeping to be ensured.<br>4. Dispensing point of flammable liquid to be kept away from turbine hall.<br>5. Seepage of oil into insulation to be avoided.<br>6. Any increase in consumption of hydrogen to be thoroughly investigated & rectified.<br>7. Adequate ventilation of TG hall.<br>8. Any welding / gas cutting work to be done only after issue of hot work permit.<br>9. Steam pipes to be thoroughly insulated. In case of oil leakage, soaked insulation to be removed and re insulation to be done. | HOD<br>(OPRN)<br>HOD<br>(OPRN)<br>HOD<br>(OPRN)<br>HOD<br>(OPRN)&<br>HOD<br>(TMD)<br>HOD<br>(OPRN)<br>HOD<br>(OPRN)<br>HOD<br>(TMD)<br>HOD<br>(OPRN)<br>HOD<br>(TMD) |  |
| 4.12    | Mineral oil filled transformers | 1. Electrical faults inside the transformers including tap changer failures.<br>2. Oil leakage from transformers.<br>3. Bushing failure<br>4. Rupture of transformer tank body caused by internal faults.                                                                                                                                                                                                                                                                       | 1. Oil leakage to be promptly attended<br>2. Vegetation rags etc. not to be allowed nearby.<br>3. Routine checking of cooling fans & pumps, operational parameters.<br>4. Oil soaking pit to be regularly inspected and cleared as necessary so as to avoid saturation.<br>5. Condition monitoring of transformers by thermo-vision cameras, acoustics methods<br>6. Maintenance of emulsifier.                                                                                                                                                                                                                                         | HOD (EM)<br>HOD (EM)<br>HOD (EM)<br>HOD (EM)<br>HOD (EM)<br>HOD (MM-OFS/CIVIL)                                                                                       |  |

| NTPC        |                          | SIMHADRI                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                         |  |
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| TITLE       |                          | FIRE PROTECTION OF POWER STATIONS                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                         |  |
| LMI No.     |                          | LMI/OGN/OPS/SYST/002/ Rev:04                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                         |  |
| S.N         | AREA                     | Fire Hazard                                                                                                                                                                                                                                                                                                                                                                                                       | Fire Prevention                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Resp.                                                                                                                                                                                                   |  |
| 4.13<br>(A) | Cable galleries / vaults | 1. Overheating of cables / cable joints up to temperature conditions high enough to bring the plasticizers of PVC compound into a volatile state.<br>2. Short – circuit creating sufficient heat.<br>3. Accumulation of coal dust, flammable debris, cardboard packages, fuel & lubricating oils.<br>4. Welding/ cutting inside cable gallery.<br>5. Heating during installation of heat shrinkable cable joints. | 1. Cable galleries / vaults to be kept clean and free of all extraneous combustible materials and not to be used as storage areas. Regular inspection of cable galleries.<br>2. Proper ventilation to be provided.<br>3. All cable entry points to cable gallery to be fire sealed.<br>4. Fire barrier walls to be provided between cable tunnels of units and fire doors to be kept closed.<br>5. Loosely laid wires to be avoided.<br>6. Lighting fittings, lighting wires to be inspected regularly.<br>7. No welding work to be done in cable galleries<br>8. Any leakage of fuel / lubricating oil into cable gallery to be immediately attended.<br>9. Any overheated joint should be repaired at the earliest.<br>10. Proper illumination in the cable gallery to be maintained.<br>11. Sprinkler system to be maintained in good condition.<br>12. Fire detection system shall be maintained & periodical inspection required. | HOD ( EM)<br><br>HOD ( EM)<br>HOD ( EM)<br>HOD (OFS) & HOD(CIVIL)<br>HOD ( EM)<br>HOD ( EM)<br>HOD ( OFS) & HOD (CIVIL)<br>HOD (MM-TMD)<br>HOD ( EM)<br>HOD ( EM)<br>HOD(OFS & HOD(CIVIL)<br>HOD ( C&I) |  |

| NTPC        |                                        | SIMHADRI                                                                                    |                                                                                                                               |                |
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| TITLE       |                                        | FIRE PROTECTION OF POWER STATIONS                                                           |                                                                                                                               |                |
| LMI No.     |                                        | LMI/OGN/OPS/SYST/002/ Rev:04                                                                |                                                                                                                               |                |
| S.N         | AREA                                   | Fire Hazard                                                                                 | Fire Prevention                                                                                                               | Responsibility |
| 4.13<br>(B) | Cables in trays in boiler area         | 1. Accumulation of coal dust, flammable debris, cardboard packages, fuel & lubricating oils | 1. Cable trays to be covered properly from top where necessary.                                                               | HOD ( EM)      |
|             |                                        | 2. Welding / cutting inside cable gallery                                                   | 2. Cable trays to be regularly inspected and cleaned. Care to be taken during gas cutting / welding near / above cable trays. | HOD ( EM)      |
|             |                                        | 3. Hot flue gas / ash from boiler furnace during furnace pressurization                     | 3. Loosely laid wires to be avoided.                                                                                          | HOD ( EM)      |
|             |                                        | 4. Heat radiation / high temperature due to vicinity of furnace, P.C. pipes, hot air ducts. | 4. Any leakage of fuel oil / lubricating oil on cables to be immediately attended.                                            | HOD ( MM-BMD)  |
| 4.13<br>(C) | Cables in trays near bottom ash hopper | 1. Accumulation of coal dust, flammable debris, cardboard packages, fuel & lubricating oils | 1. Cable trays to be covered properly from top where necessary.                                                               | HOD ( EM)      |
|             |                                        | 2. Welding / cutting inside cable gallery                                                   | 2. Cable trays to be regularly inspected and cleaned. Care to be taken during gas cutting / welding near / above cable trays. | HOD ( MM-AHM)  |
|             |                                        | 3. Hot flue gas / ash from boiler furnace pressurization                                    | 3. Loosely laid wires to be avoided.                                                                                          | HOD ( EM)      |
|             |                                        | 4. Heat radiation / high temperature due to vicinity of furnace, P.C. pipes, hot air ducts. | 4. Care to be taken during gas cutting / welding near/above cable trays.                                                      | HOD ( MM-AHM)  |
|             |                                        | 5. Hot bottom ash falling on cables during abnormal operation.                              | 5. Any leakage of fuel oil / lubricating oil on cables to be immediately attended.                                            | HOD ( MM-BMD)  |

| NTPC                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                      |                                                                                                                                                                     | SIMHADRI                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                              |
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| TITLE                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                      |                                                                                                                                                                     | FIRE PROTECTION OF POWER STATIONS                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                              |
| LMI No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                      |                                                                                                                                                                     | LMI/OGN/OPS/SYST/002/ Rev:04                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                              |
| S.N                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | AREA                                                 | Fire Hazard                                                                                                                                                         | Fire Prevention                                                                                                                                                                                                                                                                                                                                                                                                   | Responsibility                                                                                               |
| 4.14                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Switch gears                                         | 1. Insulating oil of circuit breakers, PVC cables, epoxy resin based and paxolin / melamine insulating materials, PVC wiring, solenoids and plastic encased relays. | 1. Switchgear rooms to be kept clean and free of flammable debris.<br>2. Regular inspection of switchgear rooms to be done.<br>3. Switchgear rooms not to be used as storage areas or office space.<br>4. Proper maintenance of switchgears and regular testing of protection relays.<br>5. Proper ventilation of switch gear rooms                                                                               | HOD (EM)<br><br>HOD (EM)<br><br>HOD (EM)<br><br>HOD (EM)<br><br>HOD (OPRN)& HOD (EM)                         |
| 4.15                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Diesel generators & Diesel engines driven fire pumps | 1. Leakage of fuel oil<br>2. Lubricating oil getting overheated & leaked<br>3. NRV of diesel return line to tank not working.                                       | 1. Ventilation should be ensured in the diesel engine room at high and low levels to disperse vapors and fumes.<br>2. Exhaust pipes should be suitably insulated to prevent ignition of flammable materials.<br>3. Fuel, lubricating oil leakages should be immediately attended to.<br>4. Instrumentation and controls of diesel engine to be regularly checked and maintained.<br>5. Overloading to be avoided. | HOD (OPRN)<br><br><br>HOD(OFS)<br>HOD(CIVIL)<br><br>HOD(OFS)<br>HOD(CIVIL)<br><br>HOD (EM)<br><br>HOD (OPRN) |
| <b>5.0 FIRE DETECTION AND PROTECTION SYSTEMS</b><br>NTPC- Simhadri power station is provided with fire detections systems in all hazardous areas. There are certain variations with regards to the type of such systems due to change of technology from time to time. Common types of fire detection and protections systems in NTPC stations are described in the following table for information. However, station may have to maintain the systems already installed there. |                                                      |                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                              |

| NTPC    |                                                                 | SIMHADRI                                                                                         |                                                   |         |
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| TITLE   |                                                                 | FIRE PROTECTION OF POWER STATIONS                                                                |                                                   |         |
| LMI No. |                                                                 | LMI/OGN/OPS/SYST/002/ Rev:04                                                                     |                                                   |         |
| SN      | AREA                                                            | DETECTION SYSTEM                                                                                 | PROTECTION SYSTEM                                 | REMARKS |
| 5.1     | Main plant TG Building                                          | Block                                                                                            |                                                   |         |
| a       | Unit turbine lube oil tanks, coolers and purifiers              | Quartzoid bulb type heat detectors (with hydraulic detection pipe network)                       | Automatic HVW Spray system and fire extinguishers |         |
| b       | Central lube oil tanks assembly and purifiers                   | Quartzoid bulb type heat detectors (with hydraulic detection pipe network)                       | Automatic HVW Spray system and fire extinguishers |         |
| c       | Turbine lube oil pipes(In oil canal)                            | Quartzoid bulb type heat detectors (with hydraulic detection pipe network)                       | Automatic HVW Spray system and fire extinguishers |         |
| d       | Lube oil system (tanks, piping if any) of TD – Boiler feed pump | Quartzoid bulb type heat detectors (with hydraulic detection pipe network)                       | Automatic HVW Spray system and fire extinguishers |         |
| e       | Generator seal oil system                                       | Quartzoid bulb type heat detectors (with hydraulic detection pipe network)                       | Automatic HVW Spray system and fire extinguishers |         |
| f       | Cable galleries/cable vault                                     | Linear heat sensing cable type heat detectors; ionization and photoelectric type smoke detectors | Automatic MVW Spray system and fire extinguisher  |         |

| TITLE      |                                                                                               | FIRE PROTECTION OF POWER STATIONS                                                                             |                                                                                 |                                                                                                   |
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| LMI No.    |                                                                                               | LMI/OGN/OPS/SYST/002/ Rev:04                                                                                  |                                                                                 |                                                                                                   |
| SN         | Area                                                                                          | Detection system                                                                                              | PROTECTION SYSTEM                                                               | REMARKS                                                                                           |
| <b>5.2</b> | <b>Main plant Boiler/WH</b>                                                                   | <b>RB Block</b>                                                                                               |                                                                                 |                                                                                                   |
| a          | Electro – static - precipitator                                                               | -                                                                                                             | Hydrant system, water monitors around ESPs and WHRB                             | Landing valves & individual hose boxes, hose reels                                                |
| b          | Boiler burner front                                                                           | Quartzoid bulb type heat detectors ( With pneumatic detection pipe network ) or electrical spot type detector | Automatic HVW spray system and fire extinguishers                               |                                                                                                   |
| c          | All control rooms ,MCC and switchgear rooms (ESP/VFD, Ash handling plant and other buildings) | Ionization type smoke detectors                                                                               | Fire extinguishers                                                              |                                                                                                   |
| d          | Cable galleries (ESP &VFD Building)                                                           | Linear heat sensing cable type heat detectors , ionization and photoelectric type smoke detector              | Automatic MVW spray system and fire extinguishers                               |                                                                                                   |
| e          | Boiler staircases                                                                             | -                                                                                                             | Hydrant system, water monitors on either side of boilers and fire extinguishers | Landing valves & individual hose boxes                                                            |
| f          | Coal bunkers                                                                                  | -                                                                                                             |                                                                                 | If smoke detected bunker level to be brought down by taking mill in service and fresh coal top up |
|            |                                                                                               |                                                                                                               |                                                                                 |                                                                                                   |

NTPC

SIMHADRI

TITLE

**FIRE PROTECTION OF POWER STATIONS**

LMI No.

LMI/OGN/OPS/SYST/002/ Rev:04

| S No.      | AREA                                                                                                            | DETECTION SYSTEM                                                                                                 | PROTECTION SYSTEM                                                   | REMARKS                                |
|------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------|
| <b>5.3</b> | <b>Transformer yard area</b>                                                                                    |                                                                                                                  |                                                                     |                                        |
| a          | All transformers                                                                                                | Quartzoid bulb type heat detectors ( with hydraulic / pneumatic detection pipe network)                          | Automatic HVW Spray system<br>Hydrant system and fire extinguishers | Hydrant valves & individual hose boxes |
| b          | Along the periphery                                                                                             | -                                                                                                                | Hydrant system                                                      | Hydrant valves & individual hose boxes |
| c          | DG set / Black start DG area                                                                                    | Quartzoid bulb type heat detectors ( with hydraulic / pneumatic detection pipe network)                          | Automatic HVW Spray system<br>Hydrant system and fire extinguishers |                                        |
| d          | All control rooms, MCC and switchgear rooms (compressor house, DG set area or any other local MCC/ Swgr. rooms) | Ionisation type smoke detector / Multi sensor smoke detector / photo electric smoke detector above false ceiling | fire extinguishers                                                  |                                        |

**5.4 COAL HANDLING PLANT**

|   |               |                                                                                            |                                                                                                                |                                            |
|---|---------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| a | Coal conveyor | a) LHS (Linear heat sensing cable)<br>b) RD( Infra red detector)<br>c) QB (Quartzoid bulb) | a) Hydrants/ water monitors<br>b) sprinkler system<br>solenoid manual /QB actuated/ automatic MVW spray system | Hoses to be provide in central hose houses |
|---|---------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------|



## TITLE

## FIRE PROTECTION OF POWER STATIONS

LMI No.

LMI/OGN/OPS/SYST/002/ Rev:04

|   |                                                                   |                                                                          |                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---|-------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| b | Transfer point & crusher houses                                   | Quartzoid bulb type heat detectors                                       | a)Automatic MVW Spray /sprinkler system and hydrant system (landing valves and/or water monitors)<br>a)Hydrants |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| c | Coal handling plant control rooms ,MCC and switch gear rooms      | Smoke detectors                                                          | a)Fire extinguishers<br>b)Hydrants (outside)                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| d | Cable galleries in CHP control/ switch gear rooms (if any)        | Linear heat sensing cable type heat detectors: smoke detectors           | a)Automatic MVW spray system and fire extinguishers<br>b)Hydrant (outside)                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| e | Transformers of rating 10 MVA and above within the plant premises | Quartzoid bulb type heat detectors(with hydraulic detector pipe network) | Automatic HVW spray system and fire extinguishers hydrant system                                                | Hoses to be provided in central hose houses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| f | <b>Stacker Reclaimer Machines</b>                                 | <b>Quartzoid bulb type heat detectors</b>                                | <b>Hydrants / Water Monitors<br/>Automatic MVW Spray System and Fire Extinguishers</b>                          | <b>MVW spray system shall cover both boom conveyor&amp; intermediate &amp;/or tripper conveyors and associated structure. Water for spray system shall be tapped from hydrant header running along the stockyard. For this, combination of butterfly valves &amp; quick cam couplings (SS-304construction) with one end flanged shall be provided at every 20M interval in the hydrant header running along the stock yard. Whenever, Stacker-reclaimer is in stationary condition, a flexible hose (SS-304 construction) with suitable end coupling shall be connected to quick cam coupling installed on hydrant header. The other end of flexible hose shall be permanently connected to Deluge valve."</b> |

|                       |                                             |                                                                                                                          |                                                                                      |                                             |
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| LMI No.               |                                             | LMI/OGN/OPS/SYST/002/ Rev:04                                                                                             |                                                                                      |                                             |
| 5.5 Fuel oil handling |                                             |                                                                                                                          |                                                                                      |                                             |
| a.                    | Fuel oil tanks<br>HFO/LDO                   | Linear heat sensing cable type heat detectors, Quartzoid bulb type(with pneumatic detection pipe network) heat detectors | Foam injection system and automatic/ manual MVW spray system (for uninsulated tanks) |                                             |
| b                     | Fuel oil dyke                               | -                                                                                                                        | Hydrant system(Hydrants &water monitors)<br>Foam water hydrants/monitors             | Hoses to be provide in central hose houses  |
| c                     | Fuel oil pump house equipments              | Quartzoid bulb' type heat detectors(with hydraulic detector pipe network)                                                | a)Automatic MVW/foam spray system<br>b)Fire extinguishers, foam hydrants             |                                             |
| d                     | Control rooms, MCC/switch gear rooms if any | Ionization type smoke detectors/Multi sensor smoke detector /photo electric smoke detector above false ceiling           | Fire extinguishers                                                                   |                                             |
| e                     | Transformers of rating 10MVA                | Quartzoid bulb type heat detectors(with hydraulic detector pipe network)                                                 | Automatic HVW spray system and fire extinguishers hydrant system                     | Hoses to be provided in central hose houses |

|               |                                                      |
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## 5.6 ALL OTHER OFF-SITE AREA

|    |                                                                                                          |                                                                                                                                    |                                                                                                                 |                                                |
|----|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| a  | All pump houses/<br>permanent<br>structure such as<br>WTP,PT Plant<br>etc.(other than<br>cooling towers) | -                                                                                                                                  | Hydrant<br>system(Hydr<br>ants &/ or<br>water<br>monitors) fire<br>extinguishers                                | Hoses to be provided in<br>central hose houses |
| b  | Miscellaneous<br>switchgear/MCC&<br>control rooms                                                        | Ionization type<br>smoke<br>detectors/multi<br>sensor smoke<br>detector/photo<br>electric smoke<br>detector above<br>false ceiling | Hydrant<br>system(Hydr<br>ants &/ or<br>water<br>monitors) fire<br>extinguishers                                |                                                |
| c  | Transformers of<br>rating 10 MVA and<br>above within the<br>plant premises                               | Quartzoid bulb<br>type heat<br>detectors(with<br>hydraulic<br>detector pipe<br>network)                                            | Automatic<br>HVW spray<br>system and<br>fire<br>extinguishers<br>hydrant<br>system and<br>fire<br>extinguishers | Hoses to be provided in<br>central hose houses |
| d  | Administration<br>building(periphery)                                                                    | -                                                                                                                                  | Hydrant<br>system                                                                                               | Hydrant valves & individual<br>hose boxes      |
| e. | Transformers<br>of rating<br>10MVA                                                                       | Quartzoid bulb<br>type heat<br>detectors(with<br>hydraulic<br>detector pipe<br>network)                                            | Automatic<br>HVW spray<br>system and<br>fire<br>extinguishers<br>hydrant<br>system                              | Hoses to be provided in central<br>hose houses |

| TITLE   |                                                                                       | FIRE PROTECTION OF POWER STATIONS                                                                                 |                                      |                                        |
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| LMI No. |                                                                                       | LMI/OGN/OPS/SYST/002/ Rev:04                                                                                      |                                      |                                        |
| e.      | Administration building (staircases)                                                  | -                                                                                                                 | Hydrant system and fire extinguisher | Landing valves & individual hose boxes |
| f.      | Communication building / sitcom building if any switchgear room, control/cubicle room | Ionisation type smoke detectors / multi sensor smoke detector / photo electric smoke detector above false ceiling | Fire extinguishers                   |                                        |
| g.      | Communication building / sitcom building (periphery)                                  |                                                                                                                   | Hydrant system                       | Hydrant valves & individual hose boxes |

|                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                             |                        |
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| TITLE                                                                                                                                                                                                                            | FIRE PROTECTION OF POWER STATIONS                                                                                                                                                                                                                                                                                                                                                                           |                        |
| LMI No.                                                                                                                                                                                                                          | LMI/OGN/OPS/SYST/002/ Rev:04                                                                                                                                                                                                                                                                                                                                                                                |                        |
| <b>6.0 . Inspection, Testing and Maintenance of different fire detection &amp; protection system :</b> The recommended inspection, testing and maintenance requirements of fire detection and protection systems are given below |                                                                                                                                                                                                                                                                                                                                                                                                             |                        |
| <b>a) Hydrant and fire water system:</b>                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                        |
| <b>Frequency</b>                                                                                                                                                                                                                 | <b>Checks</b>                                                                                                                                                                                                                                                                                                                                                                                               | <b>Resp.</b>           |
| Daily                                                                                                                                                                                                                            | 1 .Check and record pressure gauge reading of fire water header.                                                                                                                                                                                                                                                                                                                                            | HOD<br>(OPRN)          |
| Weekly                                                                                                                                                                                                                           | 1. Check and record discharge pressure of running electric pump and take trial of diesel driven pump ( including booster pumps) by running at least for 5 minutes weekly in general shift hours.                                                                                                                                                                                                            | HOD<br>(OPRN)          |
|                                                                                                                                                                                                                                  | 2. Checks main fire water header, piping in the pump house for any leakages.                                                                                                                                                                                                                                                                                                                                | HOD<br>(OPRN)          |
|                                                                                                                                                                                                                                  | 3. Inspect isolating valves for valve position and for any leakage                                                                                                                                                                                                                                                                                                                                          | HOD<br>(OPRN)          |
|                                                                                                                                                                                                                                  | 4. Check level and specific gravity of electrolyte in the batteries of diesel engines                                                                                                                                                                                                                                                                                                                       | HOD (EM)               |
|                                                                                                                                                                                                                                  | 5. Check battery charge panels for proper functioning                                                                                                                                                                                                                                                                                                                                                       | HOD (EM)               |
| Monthly                                                                                                                                                                                                                          | 6 .Change over to other pumps are per “Change over schedule”                                                                                                                                                                                                                                                                                                                                                | HOD<br>(OPRN)          |
|                                                                                                                                                                                                                                  | 1.Check and record pressure gauge reading of hydrant system at the highest and farthest points and other strategic points. These should be 3.5 kg/cm^2 minimum).                                                                                                                                                                                                                                            | HOD<br>(OPRN)          |
|                                                                                                                                                                                                                                  | 2.Open randomly selected valves one by one and observe flow of water for a short time. Close the valves. The whole operation of valves should be smooth and there should not be any leakage.                                                                                                                                                                                                                | HOD<br>(OPRN)          |
|                                                                                                                                                                                                                                  | 3.Run each diesel engine for about 30 minutes till the time operating parameters like exhaust temperature, closed circuit water temperature, lube oil pressure etc., are stabilized. Check whether operating parameters are within recommended range . check exhaust pipes of each diesel engine for leakage, chokage, overheating when the pump is running. Check the diesel level and top up if required. | HOD<br>(OPRN)          |
|                                                                                                                                                                                                                                  | 4.All hydrant valves and monitors to be examined systematically to ensure that all valves and spring catches are maintained in good condition along with hand –wheels, couplings, lugs etc. through CISF fire wing                                                                                                                                                                                          | HOD<br>(MM-OFS /CVL) & |
| 5. Monthly check of fire pumps control-cum-annunciation panel for proper functioning and alarm/ annunciation. One alarm to be checked randomly.                                                                                  | HOD(FM)<br>HOD(C&I)                                                                                                                                                                                                                                                                                                                                                                                         |                        |

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| Annually                                                                                        | 1. Replace a few sprinklers by open nozzles and start spray. Measure the running pressure at the hydraulic all most remote point. It should be more than 1.4kg/cm <sup>2</sup> .<br>2. Various pressure /temperature switches/gauges to be calibrated.                                                                                                                                                                   | HOD (OPRN)&HOD (MM-OFS/CIVL)<br>HOD (C&I)                                                 |
|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Every statutory outage                                                                          | Pressure vessels should be examined internally, and the opportunity taken to repaint with rust-resisting paint.                                                                                                                                                                                                                                                                                                          | HOD (MM-OFS)                                                                              |
| Every 5 years                                                                                   | Drain, clean, repair and return to service all pump suction tanks for the systems.                                                                                                                                                                                                                                                                                                                                       | HOD (MM-OFS)                                                                              |
| <b>C) Inspection, Testing and Maintenance Requirements for High and Medium Velocity Systems</b> |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                           |
| Frequency                                                                                       | Checks                                                                                                                                                                                                                                                                                                                                                                                                                   | Resp.                                                                                     |
| Weekly                                                                                          | 1. Check main water and air/water detection line pressure gauges to ensure that required pressure is maintained.<br>2. Check that sluice valves are strapped to their normal operating mode.<br>3. Check tank water level.<br>4. Check for any leakage of air/water from detection/mainlines<br>5. Check for correct position of local/manual selector switch.                                                           | HOD (OPRN)<br><br>HOD (OPRN)<br><br>HOD (PGIQ)<br><br>HOD (MM-OFS/CIVL)<br><br>HOD (OPRN) |
| Monthly                                                                                         | Transformers-Check for proper functioning of solenoid valves for remote/manual operation and local/manual operation by energizing solenoid valves but not allowing spray to take place. <b>PTW responsibility.</b>                                                                                                                                                                                                       | HOD (OPRN,EMD,C&I,OFS)                                                                    |
| Monthly                                                                                         | 1. Check for proper functioning of solenoid valves for remote/ manual operation and local/manual operation by energizing solenoid valves but not allowing spray to take place.<br>2. Check quartzoid bulbs and spray nozzles for proper positioning / orientation<br>3. Quarterly checking of Unit auxiliaries includes cable galleries, MOT,COT/DOT, seal oil, oil canal, TDBFPs, Boiler front and ESP cable galleries. | HOD (C&I)<br><br>HOD (MM-OFS/CIVL)<br><br>HOD( OPRN,C&I, OFS)                             |
| Annually                                                                                        | 1. Check all control valves, isolating valves and deluge valves.<br>2. Various pressure/temperature switches/gauges to be calibrated.<br>3. Flush detection and spray lines and clean all nozzles.                                                                                                                                                                                                                       | HOD (MM-OFS/CIVL)<br>HOD (C&I)<br>HOD (MM-OFS/CIVL)                                       |

|                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                      |  |
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| LMI No.                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                              | LMI/OGN/OPS/SYST/002/ Rev:04                                                         |  |
| Every Statutory outage                                                      | Pressure vessels should be examined internally, and the opportunity taken to repair with rust-resisting paint.<br>2. Test the deluge system by doing actual spray. Where plant operations prevent routine testing of fixed water sprays, consideration should be given to conducting a trial of the equipment during a plant overhaul.<br>3. On opportunity/shutdown, actual spray of transformers & auxiliaries shall be carried out on availability basis. | HOD (MM-OFS/CIVL)<br><br>HOD (OPRN)&<br>HOD (MM-OFS/CIVL)<br><br>HOD (OPRN,C&I, OFS) |  |
| Every 15 years                                                              | Drain, clean, repair and return to service to service all pump suction tanks for the system.                                                                                                                                                                                                                                                                                                                                                                 | HOD (MM-OFS/CIVL)                                                                    |  |
| <b>D) Inspection, Testing and Maintenance Requirements for Foam Systems</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                      |  |
| <b>Frequency</b>                                                            | <b>Checks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Resp.</b>                                                                         |  |
| Weekly                                                                      | 1. Check level of foam in tanks<br>2. Check position of valves<br>3. Check for any leakage of foam.                                                                                                                                                                                                                                                                                                                                                          | HOD (OPRN)                                                                           |  |
| Quarterly                                                                   | 1. Check yard foam spray hydrant valves for hand wheel, coupling and its lug etc through CISO fire wing.<br>2. Inspect and clean strainers if required                                                                                                                                                                                                                                                                                                       | HOD (MM-OFS/CIVL)                                                                    |  |
| 6 monthly                                                                   | Turn foam maker by 180 degrees and open the foam line for 5 minutes. Observe the quality of foam being made and the rate at which foam is being sucked by pump                                                                                                                                                                                                                                                                                               | HOD (MM-OFS/CIVL)                                                                    |  |
| Biennial                                                                    | 1. Check the quality of foam and replace/top up if necessary.<br>2. Inspect internal surface of foam vessel and the opportunity taken to repaint with rust –resisting paint.<br>3. clean and service all valves and pumps.<br>4. various pressure / temperatures switches/gauges to be calibrated.                                                                                                                                                           | HOD (MM-OFS/CIVL)<br>HOD (MM-OFS/CIVL)<br>HOD (MM-OFS/CIVL)<br>HOD (C&I)             |  |

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**E) Inspection, Testing and Maintenance Requirements for INERGEN system**

| Frequency     | Checks                                                                                                                                                | Resp.                                                  |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Quarterly     | 1. Check pressure gauges readings ,check the setting.<br>2. Check the functioning of C&I instruments.                                                 | HOD ( OPRN)<br>HOD (C&I)                               |
| 6 monthly     | Examine the cylinders content level and replace the cylinders if necessary.                                                                           | HOD (MM-OFS/CIVL)                                      |
| Annually      | 1.Check that correct no. of cylinders is in service.<br>2. Calibration of C&I instruments<br>3. Simulation checking of st1 &2 Intergeren /NN100system | HOD ( OPRN)&<br>HOD (MM-OFS/CIVL)<br>HOD (C&I)&<br>HOD |
| Every 5 years | <b>Actual Discharge Test to be done in smallest risk area.</b><br><b>Hydro test of Cylinders is to be done</b>                                        | <b>HOD ( OPRN)</b>                                     |

**F) Inspection, Testing and Maintenance Requirements for Detection Systems**

| Frequency | Checks                                                                                                                                                                                                                                | Resp.     |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Quarterly | 1. Functional test of detection system if required .In each area 2 detectors selected randomly will be checked.<br>2. Inspect physical position of LHS cable other detection system hardware for any damage and testing by simulation | HOD (C&I) |
| 6 Monthly | 1. Functional test of detection system<br>2. Check and test all control and annunciation panels.                                                                                                                                      | HOD (C&I) |
| Annually  | Check the system integrity.                                                                                                                                                                                                           | HOD (C&I) |

**G) NIFPS for Transformers**

**For Nitrogen Injection Fire Protection System (NIFPS) checks to be done as per OEM recommendations and Hydro test of Cylinders is to be done once in 5Y.**

**7.EXCEPTION REPORTING**

The exception encountered during the inspection are to the reported as per the following format by HOD'S of MM-OFS/CIVL/EMD/C&I/OPERATION/CHP

| Description of exception | Area | Location of exception | Remarks |
|--------------------------|------|-----------------------|---------|
|                          |      |                       |         |



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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                   |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                   |  |
| <b>8.0 RESPONSIBILITIES</b><br>The responsibility of carrying out different tasks in relation to preventive measures, inspection and testing of above systems as detailed in the LMI are with HOD-Operation, HOD(MMOFS/CIVIL),HOD(MM-BMD), HOD-FM,HOD-C&I and HOD- EM. HOD (IT).                                                                                                                                                                                   |  |                                   |  |
| <b>9.0 RECORD KEEPING</b><br>The records to be maintained under this LMI include: <ul style="list-style-type: none"> <li>a) Compliance record of the activities detailed under this LMI <b>in Offsite areas and CHP area will be maintained by the respective departments .</b></li> <li>b) Exception reporting , if any.</li> </ul> The above records are to be maintained by HOD-Operation ,HOD(MM-OFS/CIVIL),HOD(MM- BMD),HOD-(FM),HOD-C&I and HOD(EM),HOD(IT). |  |                                   |  |
| <b>10.0 REVIEW:</b> This document shall be reviewed by HOP, Simhadri on biennial basis.                                                                                                                                                                                                                                                                                                                                                                            |  |                                   |  |

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| <b>10.DISTRIBUTION LIST:-</b> A soft copy of the LMI shall be uploaded in the EEMG dept. website on Simhadri intranet LAN by the EEMG dept.(Password Protected) The uploaded soft copy of the document is to be considered as Controlled Copy. |                                          |

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| TITLE                                    |        |            | FIRE PROTECTION OF POWERSTATIONS                                                                                                                                                                                                                                                                                                                                                                                                 |             |  |
| LMI No.                                  |        |            | LMI/OGN/OPS/SYST/002/ Rev:04                                                                                                                                                                                                                                                                                                                                                                                                     |             |  |
| 11.REVISION INDEX (for changes/revision) |        |            |                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |  |
| SI.NO                                    | REV.NO | DATE       | DETAILS                                                                                                                                                                                                                                                                                                                                                                                                                          | APPROVED BY |  |
| 01                                       | 01     | 15.05.2013 | 1. Change in department Name and designations.<br>2. Frequency of checking and recording discharge pressure of running electric pump and trial run of diesel driven pump (including booster pumps)by running at least for 5 minutes changed from daily to weekly                                                                                                                                                                 |             |  |
|                                          | 02     | 15.05.2020 | 1. Change in department Name, designations.<br>2. Clause -7.5Coalfeeders Fire hazards Point no4&5, preventive action written in detail.<br>3. Clause-7.12 Mineral oil filled Transformer Fire prevention measure column Point No 6 added with mentioning of Responsibility after discussions with EMD.<br>4.7.13A Cable Galleries / Vaults-in Fire Prevention columnPointNo-11&12 added by HOD —EM & Mentioned responsibilities. |             |  |
|                                          | 03     | 06.08.2022 | Revised as per OGN-OPS-SYST-002, REV.1 guidelines & change in department name, designations.                                                                                                                                                                                                                                                                                                                                     |             |  |
|                                          | 04     | 11.03.2025 | Revised as per COS-ISO-00-OGN/OPS/SYST/002<br>Rev. No.: 2, Dec. 2024 and marked in BOLD letters                                                                                                                                                                                                                                                                                                                                  |             |  |

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| COMPLIANCE RECORD              |          |                                   |  |
| SYSTEM / EQUIPMENT DESCRIPTION | ACTIVITY | COMPLIANCE                        |  |
|                                |          |                                   |  |

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| ANNEXURES |                                   |                       |  |
| SL.NO     | DOCUMENT NAME                     | RESPONSIBILITY CENTER |  |
|           |                                   |                       |  |

